

MATERNOVA

Empowering ASHA Workers with AI-Driven Offline Rural Care

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Problem Statement



Manual registers & data loss



Poor Internet connectivity



Delayed maternal and elderly & child care



Fragmented medical history

Objectives

- Digitalize rural health records
- AI-based test & diet recommendations
- Reduce maternal, infant & aged person mortality

Our Prototype

Maternova

hemata

Dashboard

Emergency Alerts

Vaccination Records

Treatment Records

Meal Menu

Nutrition Guide

Government Funding

Namaste, hemata!

4 Active Alerts

12 Vaccinations

5 Meal Plans

P0.8L Funding Received

Quick Actions

Emergency

Vaccinations

Meal Menu

Nutrition

Recent Alerts

Namrata's Fever

Swati's Cough

Ramish's Vomiting

Pregnant Patient 1

Iron-Rich Breakfast Bowl

Protein-Rich Thali

Light Nutritious Dinner

Pregnant Patient 2

Calcium Boost Breakfast

Healthy Evening Snack

Beneficiary Name *

Patient name

Category *

Pregnant Woman

Treatment Type

e.g., Prenatal Care, Chronic Care

Diagnosis *

Medical diagnosis

Prescription

Medicines and dosage

Date *

dd-mm-yyyy

Doctor Name

Treating doctor

Notes

Follow-up instructions, etc.

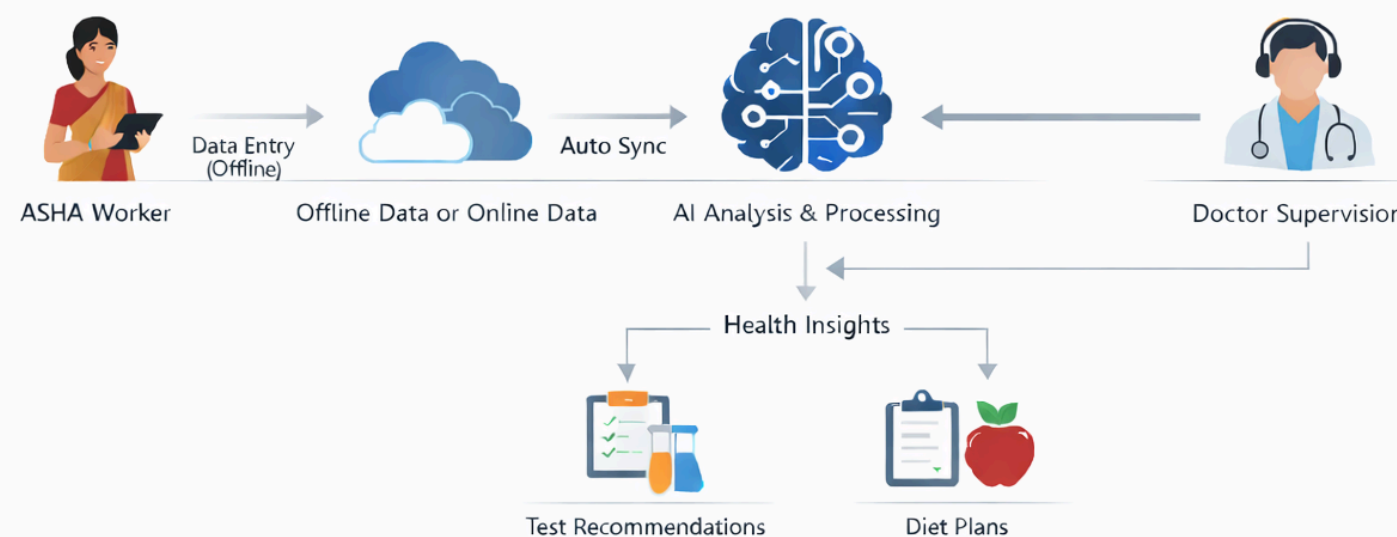
Capture Photo (Optional)

Take Photo

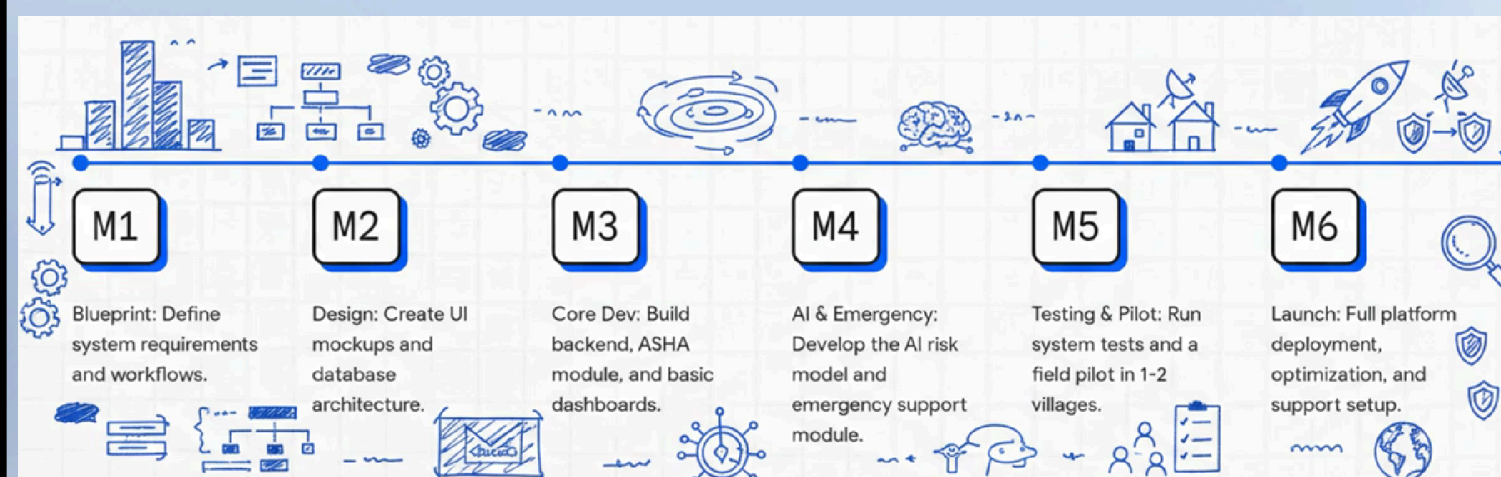
Proposed Solutions

- Offline-first ASHA mobile app
- Secure digital health profiles
- AI analysis under doctor supervision
- Auto-sync when internet available
- Emergency phone network alert

System Workflow



Implementation



References : WHO, UNICEF, NFHS, NACDA etc

<https://drive.google.com/drive/folders/1ftKgKk0unU5DJew4aP4tjIW9V3lPuhtY>

Impact

Measurable Impact & Feasibility



Missed Checkups
30-40%
Reduction



Early Risk Detection
25-35%
Improvement



Emergency Guidance Speed
50-60%
Faster



Record Keeping Efficiency
40-50%
Reduction in manual time

Scalability & Sustainability

- Integrates with PHCs and hospitals
- Government scheme compatible
- Low operational costs
- Minimal internet dependency

Future Scope

- Disease outbreak prediction
- Nationwide implementation
- Telemedicine & Specialist connectivity
- Integration with universal rural health inclusion and expansion

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